

Submission Summary

Conference Name

VI International Congress of Interdisciplinary Research #LoHacemosxTIC

Paper ID

14

Paper Title

Development of a Caterpillar Robot for Agricultural Environments with ROS2

Abstract

Developed countries are investing billions of dollars in precision agriculture and robotics to reduce the environmental impact of agricultural production processes through sustainable production. The above aims to address sustainable development goals such as i) zero hunger, ii) clean water and sanitation, and iii) industry, innovation, and infrastructure. Among the tasks that can be automated in crops such as bananas, corn, and coffee are phytopathological analysis, crop mapping, precise pesticide application, and deforestation control. Precision agriculture and agricultural robotics enable the automation of repetitive tasks, enabling continuous operations and reallocation of human resources to more strategic areas of agricultural management. However, Colombian agricultural technology lags behind that of countries such as the United States and the Netherlands. Therefore, this article presents the development of a tracked mobile robot, manufactured using 3D printing, designed to perform autonomous navigation tasks through crop rows with vegetation barriers. This mobile platform features a perception system based on a LiDAR sensor that measures the robot's lateral distance from the vegetation barrier, an inertial measurement unit for attitude estimation, and encoders for speed control and movement tracking. This article also presents preliminary results for establishing process communication between the robot's perception, processing, and control modules using ROS2.

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Submission Files

Resumen_GersonCruz_AndresJutinico.pdf (397.4 Kb, 2026-05-14 08:53:03 COT)

Submission Questions Response

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Technology and artificial intelligence

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